Universität Zürich UZH

GEO877

Spatial Algorithm

02. Centroid and Area

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Feedback/solution for your practical 1

- Integrating Haversine distance
- Calculating variance
- How to make your class Point into a "python package"
- Implement Welford's algorithm (optional) for efficiency measure (and I will show you another method – vectorisation, which might boost up the efficiency even further)

Recap

Basic elements of (spatial) data

- **Point** in coordinates (x,y) assuming Cartesian space
- **PointGroup** as a collection of points
- **Segment** as a finite line with a start and an end point
  - Segments are bounded at both ends
  - Rays have a start but no end
  - Lines have neither a start nor an end
- **Polygon** with an ordered set of point coordinates

- We will not cover raster, but can be implemented for your own code if necessary

PointGroup class

- Thematically associated group of points (ex. Trees of Canton X and Canton Y)

- Property:
  - A collection of Points, without specific spatial ordering

- Behaviour:
  - Minimum or maximum point of a group
  - Area to enclose the point group *(lectures to follow)*
    - Bounding box
    - Convex hull
  - **Size (area) and centroid** of point group

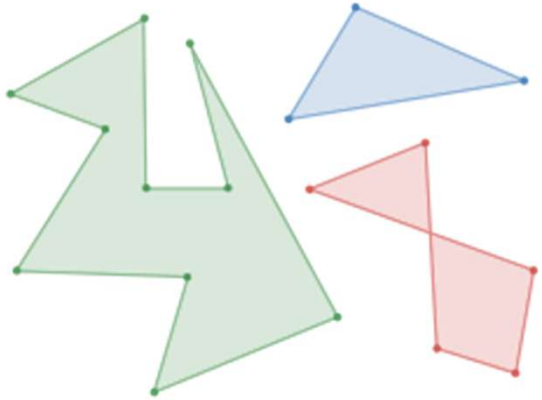
`Polygon` class

- A series of connected segments – straight lines between points – connecting back to the start to form a closed figure
- Property:
  - An ordered set of points where the last == the first
  - **Size (area), centroid**
- Behaviour:
  - Equality: we can test if two or more polygons are the same
  - Intersection: we can divide the polygon with segment(s) to create new (sub)polygons
  - Adjacency: we can test if two or more polygons share a border (i.e. segment)
  - Containment: we can test if a point is located within a polygon

Creating a polygon out of point group

Definition of polygon:
a series of connected segments – straight, limited length of lines between points – to form a closed figure

- How to create a polygon?
  - No duplicates of points, except for first and last
  - Polygon should be closed
  - No self-intersection
  - No holes or islands (e.g. donut)

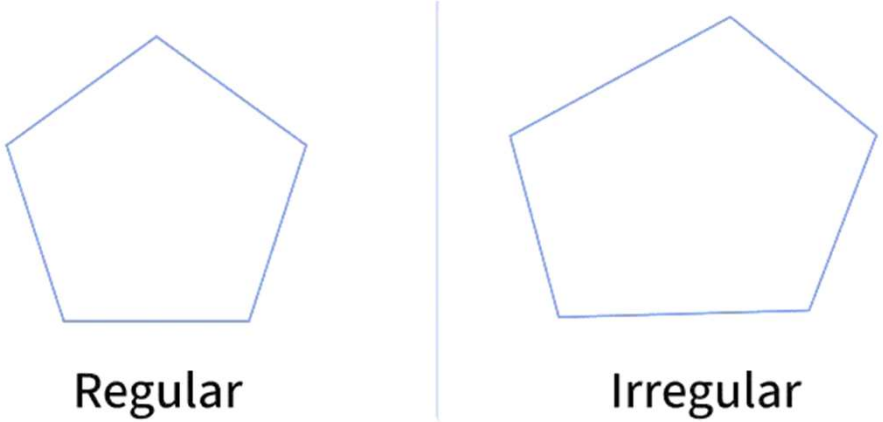


Green and blue are simple polygon; red, on the other hand, is a self-intersecting polygon (i.e. not simple)

https://en.wikipedia.org/wiki/Simple_polygon

Polygon Area

- When it comes to regular polygons, we know the 'equations' for calculating the area;
- But then how we should handle the area of irregular polygons?



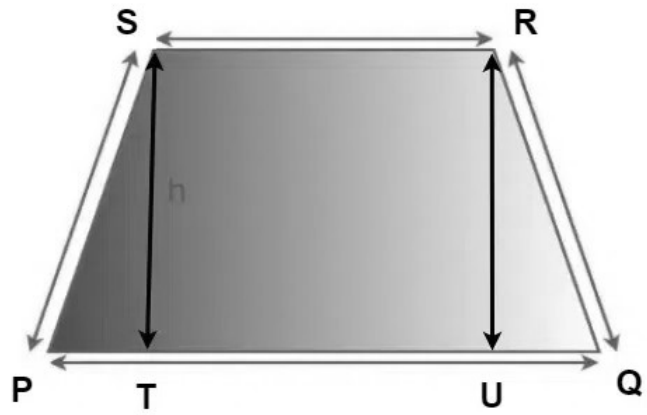
For the equations for "regular" polygons → https://www.geeksforgeeks.org/maths/polygon/

Trapezoid

One idea of calculating the area of polygon is a **trapezoid** method, i.e. summing up the area of all trapezoids out of a polygon

Properties of a trapezoid

- There is only one pair of parallel sides (otherwise, it is not a trapezoid)
- Sum of all angles of a trapezoid is 360º
- Median is parallel to both the bases and is the average length of the bases in a trapezoid

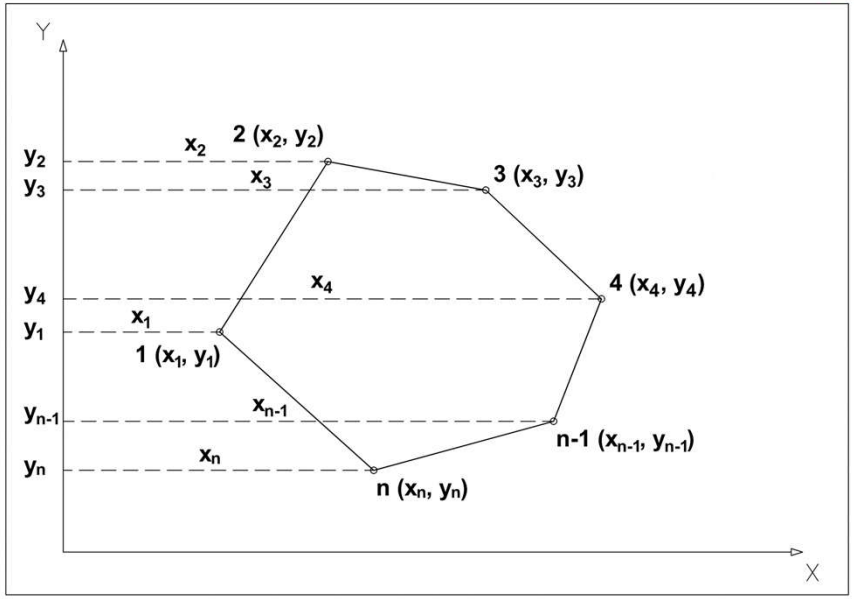


More about trapezoid → https://www.geeksforgeeks.org/maths/trapezoid-formula/ or https://www.geeksforgeeks.org/maths/area-of-trapezoid/

Polygon Area

One idea of calculating the area of polygon is a **trapezoid** method, i.e. summing up the area of all trapezoids of a pair of points in the PointGroup. Also called as the "shoelace method with geometric decomposition"

It is particularly useful for irregular polygons, but works for regular polygons too



More about A_i in the equation → https://www.geeksforgeeks.org/maths/trapezoid-formula/ or https://www.geeksforgeeks.org/maths/area-of-trapezoid/

The video comes from https://www.thecivilengineer.org/education/professional-examinations-preparation/calculation-examples/calculation-example-three-point-resection

Polygon Area

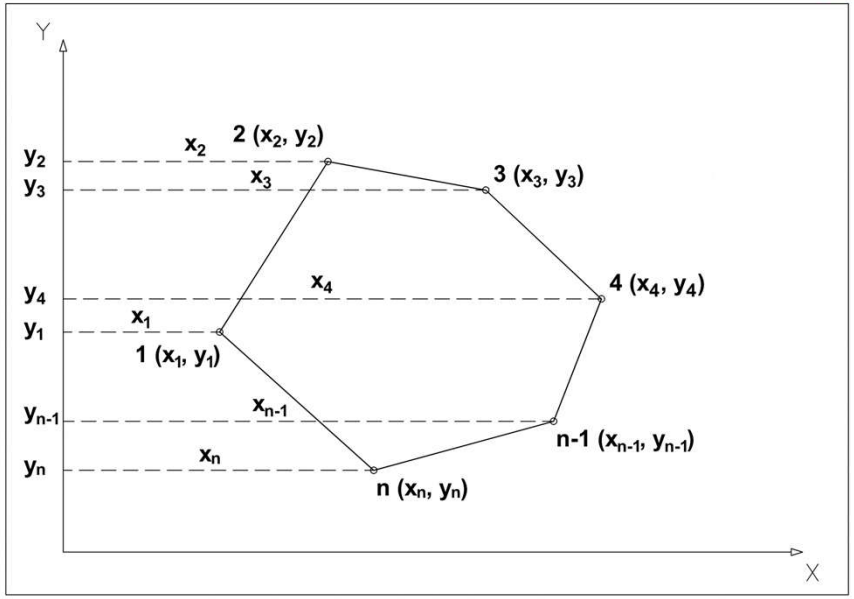
When the polygon vertices are given like this:

$$(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$$

The area of one trapezoid is

$$\boldsymbol{A_i = (x_i + x_{i+1}) * (y_i - y_{i+1}) / 2}$$

(this equation is using the y-axis for projection, i.e. horizontal dotted line)



This area is **signed** (in terms of y here):

- On y-axis: bottom → up makes negative (red)
- On y-axis: up → bottom makes positive (green)

In the end, a complete area of a polygon **A =** $\left| \sum_{i=1}^{n} \boldsymbol{A_i} \right|$

More about A_i in the equation → https://www.geeksforgeeks.org/maths/trapezoid-formula/ or https://www.geeksforgeeks.org/maths/area-of-trapezoid/

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Polygon Area

For this area equation, there are two ways to express. They look different but in fact they are the same; when you spread out the Trapezoid equation, and as i continues, $x_{i+1}y_{i+1}$ and $x_i y_i$ will be canceled out in the following iteration, and then it will end up look like the equation on the right

Trapezoid method

$$A = \frac{1}{2}\sum_{i=1}^{n} (x_{i+1} + x_i) * (y_i - y_{i+1})$$

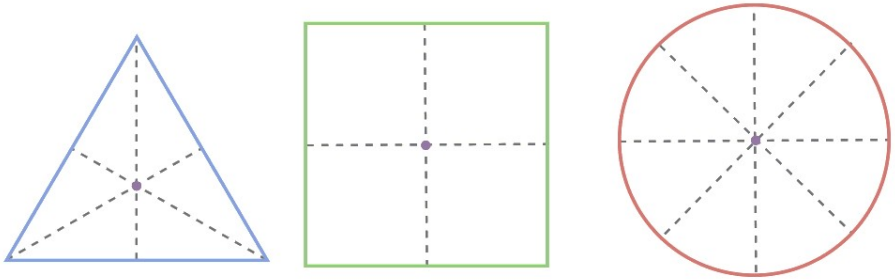
Cross product of triangle method

$$A = \frac{1}{2}\sum_{i=1}^{n} (x_{i+1}y_i - x_i y_{i+1})$$

Polygon Centroid

- Centroid operations is to identify the center of geographic objects. Like statistical measures of central tendency, and there are many ways to define the geographic center of an object
- The most commonly used centroid operation is the **geometric centroid**, i.e. the arithmetic mean position of all the points in a shape (in other words, "center of mass" or the "center of gravity")

- Why can't we just calculate the 'mean' point of all points?



https://www.geeksforgeeks.org/maths/centroid/

For more information about mean centre and centroid, go to https://www.spatialanalysisonline.com/HTML/centroids_and_centers.htm

Polygon Centroid

A trapezoid method

- Centroid of a polygon = **area-weighted** average of the centroids of all subset pieces of a polygon (e.g. trapezoids)

Area

$$A = \frac{1}{2}\sum_{i=1}^{n}(x_i y_{i+1} - x_{i+1} y_i)$$

(note: the equation itself is the same as we have seen before, but here, the equation sets x-axis as a projection ground so that x and y are swapped)

Centroid $\bar{x}$ and $\bar{y}$

$$\bar{x} = \frac{1}{6A}\sum_{i=1}^{n}(x_i + x_{i+1})\,(x_i y_{i+1} - x_{i+1} y_i)$$

Sum of area-weighted X mean

$$\bar{y} = \frac{1}{6A}\sum_{i=1}^{n}(y_i + y_{i+1})\,(x_i y_{i+1} - x_{i+1} y_i)$$

Sum of area-weighted Y mean

1/2 from shoelace method, 1/3 from triangle centroid

https://www.geeksforgeeks.org/maths/centroid-of-a-trapezoid-formula/

Practical exercise 2:

- I will show you how to create a polygon out of a PointGroup (together)
  - (Luckily, we don't need to "order" the points as it is well-prepared)
  - We should test if the polygon is closed: how do we determine that?
  - We don't need duplicate points, why?

- For your practical, you will try to add the code to calculate area and centroid in the existing Polygon class I prepared
- Play this function with a new dataset

➢ As always, we will discuss your solutions at the beginning of next class

Your group is ready

Your group is ready

- You will find your teammates on Teams
- You will have access to your private channels on Teams shortly

Your first group work will be a paper review presentation – see the next slide

Paper review presentation

10-minute presentation per group; and **5-minute** all-together discussion

Everybody on **March 26th**

The presentation should include:

1. What **research questions** were posed by the authors?
2. What **data** were used? How were the data **structured** and **organised**?
3. What **tasks – spatial questions –** were applied to carry out the analysis?
4. If you **replicate** this study on your *own*, how would you define research question(s), data source(s), and spatial algorithm(s)?